Spring 2026 ECE 445 Team Contract

Instructions: The content of this document should be specific to your goals and needs. Ideas for the content of each section are provided as suggestions.

Project No. and Name	**No 93: Dynamic Violin Fingerboard Attachment**
Member Name, netID	Adrian Ignaci, aigna3
Member Name, netID	Kamil Waz, kwaz2
Member Name, netID	Sophia Wilhelm, sophia16

ECE 445 is a project-based course. The course includes both team and individual grades. Project teammates generally all get the same grade for team assignments based on the expectation that all team members do their fair share of the work involved. The purpose of this contract is to lay out the tasks needed for the successful completion of the project and distribute them in a fair and efficient way to the team members. It will also discuss how the teammates will work together during the project and address any issues that come up. A contract that promotes good teamwork that leads to a successful project should:

- Acknowledge that each team member has commitments and responsibilities outside of ECE 445
- Encourage open communication about challenges that team members are facing, both in and out of ECE 445
- Give team members the benefit of the doubt and the opportunity to explain themselves when something goes wrong and resist jumping to judgement

Project Description:

The dynamic violin fingerboard aids self-taught violinists in learning correct finger placements and rhythms. Users upload a MIDI file for a piece they would like to learn and configure the piece settings. A dynamic LED display laid on top of the fingerboard then illuminates the locations where the user should place their fingers at the appropriate times. Potentiometers measure the position of a finger placed along any of the 4 paths (strings) on the fingerboard. These measurements are turned into an accuracy metric that is provided to the user via an LCD display. This allows us to collect information on how accurate the placement is, rather than a simple yes or no as to whether they play the right note.

Project Goals:

This project will be considered successful if:

- The LEDs on the fingerboard turn on independently from each other in response to a signal from the control unit.
- A MIDI file less than 1 MB can be uploaded to the control unit and sets the control sequence for the fingerboard LEDs.
- Potentiometer membrane can detect finger placement within 2 mm and measurements are converted to an accuracy score that is provided to the user at the end of the piece.

If time permits, the following features will be added:

- Real-time accuracy feedback.
- Multiple songs can be uploaded to the controller at once and the user can choose which to play.

Expectations (ground rules) for each member:

- Discord will be the primary method of communication. Each member will acknowledge messages sent within 24 hours.
- Members will (if reasonably possible) respond to feedback requests within 24 hours.
    - Members will give notice of periods of time in which they will not be able to respond to requests.
    - Members will provide constructive feedback upon request
    - Members understand that feedback is not personal criticism
- Each member is expected to know when overall project deadlines are.
- It is understood that members have responsibilities external to this project that may prevent them from working consistently. However, when there is a large deadline imminent (within 2 days), members will treat this project with high priority.
- Members will show progress or effort towards completing their assigned tasks each week.
- Members will participate actively: show initiative in identifying ways to move forward.

Roles:

All members have the same major (CompE) and thus have overlapping strengths. Thus, tasks are split primarily by interest.

Initial task split: Kamil handles physical design, Adrian and Sophia split the code base.

Kamil: in possession of the physical violin used for testing, so will handle tasks that require specific knowledge of the instrument.

Sophia: has demonstrated skill in soldering. Also code-focused.

Adrian: Code-focused.

Project Meeting Time(s):

TA meetings: Thursday 4:00pm (Teams)

Design meetings: Tuesday 4-6 (ECEB)

Lab meetings: Wednesday as needed (ECEB)

Emergency meetings: Friday mornings (discord)

Additional meetings scheduled as needed

Agenda:

Meeting minutes will be kept in a google document. Contents: date, time, location, attendees, agenda, minutes, objectives for next meeting, upcoming deadlines, date of next meeting. The document is shared with all members and should be uploaded after each meeting to individual member's lab notebooks. Sophia will be the primary record keeper, but other members are invited to add to the records as desired. The meeting minutes will help the team keep the project on track throughout the semester.

Larger decisions will be decided by consensus vote. Votes may happen during meetings or asynchronously over discord.

Process and penalties for dealing with team issues:

If a team member is falling behind on a critical task, that task is reassigned to another group member and the tardy member is assigned a less important task of comparable workload.

If someone is unresponsive on discord, initial escalation method will be attempting contact via school email and copying the TA.

If a member does not show up to meetings, the other members will reach out on discord. If they are unresponsive, the matter will be escalated to email.

End-of-term agreement on using final peer assessment for grade adjustment:

We feel that the contents of this contract are intended to guide the organization of the project going forward but are not specific enough to treat as a binding document. Unforeseen circumstances may affect the division of labor amongst members, so we mutually agree to act in good faith to help the project succeed to the best of our individual abilities.

Signatures: Iterate on this document until everyone is comfortable with its contents and signs (it is okay to type your printed name as your digital signature).

I affirm that I participated in generating this team charter and that I will abide by its contents to the best of my ability. Furthermore, I understand that failure to meet the expectations expressed here can lead to the stated consequences.

netID: aigna3_____ (digital) Signature: Adrian Ignaci__ Date: 2/20/2026

netID: kwaz2_____ (digital) Signature: Kamil Waz Date: 2/20/2026

netID: sophia16_____ (digital) Signature: Sophia Wilhelm Date: 2/20/2026